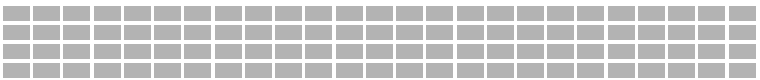
LM-head Intermediate Tensor Lifetime

Same scoring contract; different peak allocation

A0 Dense Full readout: materialize $N \times V$ logits before reduction.

$\mathbf{H}[N, d] \longrightarrow \text{LMHead} \longrightarrow \mathbf{Z}[N, V]$



Complete $[N, V]$ intermediate lives through reduction

A1 Chunked Per-tile readout: normalize and discard each $B \times V$ tile.

$\mathbf{H}_i[B, d] \longrightarrow \text{LMHead} \longrightarrow \mathbf{Z}_i[B, V]$

Per tile: LSE / gather / Top-K → write $[N, K]$ → discard $\mathbf{Z}_i$

Dense temporary Logits: $O(NV)$

Chunk temporary: $O(BV)$; compact outputs: $O(NK)$

Measured (N=8192, K=32, B=256): PyTorch peak allocated 3596.6 → 681.6 MiB (−81.05%)

Transformer hidden states, model/FSDP residency, and backend workspaces are excluded from these simplified expressions.